

ANUBHAV AGRAWAL

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SUMMARY

Machine Learning Engineer with **5+ years** of experience — **3+** of them full-time — building AI systems end-to-end: model R&D, real-time inference, *and* the production platform around them.

TECHNICAL SKILLS

Languages & Tools: Python, Git, Linux, L^AT_EX

ML & Deep Learning: PyTorch, TensorFlow, OpenCV, Scikit-Learn, XGBoost, NumPy, Pandas

LLMs & RAG: OpenAI API, LangGraph, HuggingFace, Vector Search, Cross-Encoder Re-ranking, MongoDB

Inference & Serving: TensorRT, ONNX, INT8/FP16 Quantization, FastAPI, mlx-lm, llama.cpp

Platform & MLOps: AWS (EC2), Docker, CI/CD, OpenTelemetry, Grafana LGTM (Loki, Tempo, Mimir)

EXPERIENCE

Griphic

Delhi

AI Engineer II

Jan 2026 – Present

- Built the adaptive-intelligence core of an AI interviewer: a stateful LangGraph agent that scores responses, adapts difficulty in real time, and conditions every follow-up on retrieved context.
- Built the RAG pipeline — 3 parallel retrieval paths, *hybrid keyword + dense search*, *cross-encoder re-ranking* over 512-d embeddings — at **0.95 Recall@5** across **10K+ documents**.
- Cut query latency **2.5s** → **0.8s** (68%) with context fusion and async orchestration; raised domain-query **F1** by **42%** on a held-out eval set; serves **500+ users at 99.9% uptime**.
- Deployed the production system end-to-end — Python/FastAPI on AWS EC2, MongoDB, OpenAI APIs — with a CI/CD pipeline of automated Docker builds from commit to deploy.
- Set up observability: an OpenTelemetry Collector feeding logs, traces, and metrics to the Grafana LGTM stack (Loki, Tempo, Mimir), plus LLM token-tracking and cost dashboards for per-feature spend.
- Wrote the evaluation suite — recall, faithfulness, answer-relevance — and citation-grounded LLM guardrails that keep every answer traceable to source documents.

Innvolution Healthcare Pvt. Ltd.

Bangalore

Machine Learning Engineer & Team Lead

Sep 2023 – Dec 2025

- Took AIMAG, an X-ray super-resolution & denoising product at **\$200K+ ARR**, from dataset curation through model R&D, quantization, and release to clinical trials.
- Served the model in real time: PyTorch → ONNX → TensorRT with INT8/FP16 quantization behind a FastAPI runtime — **600+ FPS** on an RTX 4090, a **70% latency cut**, <5% quality loss, **35% lower VRAM**.
- Automated the QCA pipeline (segmentation → calibration → prediction) to finish in **<4 s** with a fine-tuned U-Net++ (ResNet50 + attention) at **0.94 Dice**; deployed for live cardiac-catheterization analysis.
- Co-invented **3 patent applications** in X-ray image enhancement: single-image blind super-resolution, DSA roadmap generation, and real-time fluoroscopy enhancement.
- Led the ML team, owning roadmap, review standards, and delivery from research prototypes through production deployment.

Image Quality Engineer Intern

Nov 2022 – May 2023

- Built motion-compensated stitching & denoising for DSA sequences: **20% clarity gain** at <2-pixel alignment error.

INTERNSHIPS

ISRO-SAC: NDVI time-series forecasting on AWiFS, MODIS, Sentinel

Jun 2022 – May 2023

IIIT-NR: Smoking & spitting detection + dataset (ResNet18, transfer learning) for CCTV

May – Jun 2022

PROJECTS

Inference Lab — LLM Serving Benchmarks on Apple Silicon

[GitHub](#) | [Write-up](#)

Reproducible benchmark suite: mlx-lm vs llama.cpp under a single fairness contract

2026

- Benchmarked mlx-lm vs llama-cpp on **Qwen2.5-7B-Instruct** (4-bit) under identical conditions — fixed prompts, greedy decoding, matched budgets — measuring latency, throughput, and memory: **45–56 tok/s** single-stream on a MacBook Pro M4 Pro (24 GB unified memory).
- Found why throughput stays **flat from 1 to 32 concurrent users**: requests serialize through a per-instance lock inside each serving engine.
- Showed the limit is memory bandwidth: a multiprocessing K-instance dispatcher gains only **+9% for 3× the memory**, because decoding already saturates the M4 Pro's ~273 GB/s memory bus.

EDUCATION

Bhilai Institute of Technology, Durg — B.Tech in Computer Science Engineering

2019 – 2023